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Division of Dockets Management (HFA-305)
Food and Drug Administration
5630 Fishers Lane, Rm. 1061
Rockville, MD 20852

In Re: Docket No. FDA-2015-D-3719-0018, Draft Guidances Relating to the Regulation of Human Cells, Tissues, and Cellular and Tissue-Based Products; Rescheduling of Public Hearing; Request for Comments

Submitted electronically at www.regulations.gov

Dear Madams and Sirs:

The American Association of Tissue Banks (AATB) and the American Association of Tissue Bank's Tissue Policy Group, LLC (AATB TPG) submit these comments related to the series of draft guidances relating to the regulation of human cells, tissues, and cellular and tissue-based products (HCT/Ps).

The AATB is a professional, non-profit, scientific and educational organization. It is the only national tissue banking organization in the United States, and its membership totals more than 125 accredited tissue banks and 650 individual members. These banks recover tissue from more than 30,000 donors and distribute in excess of two and a half million allografts for more than one million tissue transplants performed annually in the U.S. More than 90 percent of the human tissue distributed for these transplants comes from AATB-accredited tissue banks.

The AATB's Tissue Policy Group (TPG), LLC (AATB TPG or TPG) includes Chief Executive Officers and senior regulatory personnel from U.S. tissue banks that process donated human tissue. The purpose of the TPG is to drive public policy in furtherance of the adoption of laws and regulations that foster the safety, quality and availability of donated tissue. The TPG's membership is responsible for the vast majority of tissue available for transplantation within the U.S.

As part of our overall discussion with the Food and Drug Administration (FDA or agency), the AATB and the TPG have requested that the FDA provide more certainty, not less, with regard to the regulation of HCT/Ps. By doing so, the AATB and the TPG believe that there will be further innovation and improved patient care. The Association and the TPG have requested, and will continue to request, clear guidance related to the current good tissue practices and manufacturing arrangements.¹

¹ See Guidance for Industry: Compliance with 21 CFR Part 1271.150(c)(1) – Manufacturing Arrangements (September 2006), *available at*

Due to that keen interest, the TPG previously submitted a document detailing what should be included in the draft homologous use guidance document prior to the FDA's publication of its own guidance document. Building upon that effort, the AATB and the TPG now submits a separate guidance document recommendation (attached) titled "Appropriate Analysis, Characterization and Assessment to Ascertain Minimal Manipulation and Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps)." We developed this draft guidance document recommendation to provide the FDA with the AATB's and the TPG's assessment of the need for a separate guidance document examining issues related to analysis, characterization and assessment of Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps) and the relationship of that analysis, characterization and assessment to minimal manipulation and homologous use for HCT/Ps.

This guidance document recommendation expands upon the previously submitted homologous use guidance document recommendation by adding new, discrete concepts. Namely, as the title suggests, the main feature of this guidance document recommendation is to provide a framework for the appropriate analysis, characterization and assessment of HCT/Ps based on the manufacturer's objective intent. This document further details key linkages between core regulatory concepts drawing a clear regulatory link between the manufacturer's objective intent; the homologous use; the original, relevant characteristics; and the appropriate analysis, characterization, and assessment. Finally, it also includes recommendations for HCT/P flow diagrams, given the need for additional clarity in this area.

We hope that you will find this guidance document recommendation useful in your deliberations related to the four draft FDA guidance documents related to HCT/Ps. The AATB and the TPG stand ready and willing to assist the FDA with its deliberations in any way that the FDA deems appropriate.

Respectfully,



Frank Wilton
President & CEO
American Association of Tissue Banks



Thomas A. Cicyota
Chair
Tissue Policy Group

Attachment: TPG's Guidance Document Recommendation Titled "Appropriate Analysis, Characterization and Assessment to Ascertain Minimal Manipulation and Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps)."



Guidance Document Recommendation

**Appropriate Analysis, Characterization and Assessment to Ascertain
Minimal Manipulation and Homologous Use of
Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps)**

All requests and comments should be addressed to:

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Introduction

We, the American Association of Tissue Banks (AATB) Tissue Policy Group (TPG) or AATB TPG, developed this draft guidance document recommendation to provide the Food and Drug Administration (FDA or Agency) with AATB TPG's assessment of the need for a separate guidance document examining issues related to analysis, characterization and assessment of Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps) and the relationship of that analysis, characterization and assessment to minimal manipulation and homologous use of HCT/Ps.

This guidance document expands upon the previously submitted homologous use guidance document recommendation by adding new, discrete concepts. Namely, as the title suggests, the main feature of this guidance document recommendation is to provide a framework for the appropriate analysis, characterization and assessment of HCT/Ps based on the manufacturer's objective intent. This document further details key linkages between core regulatory concepts drawing a clear regulatory link between the manufacturer's objective intent; the homologous use; the original, relevant characteristics; and the appropriate analysis, characterization, and assessment. Finally, it also includes recommendations for HCT/P flow diagrams, given the need for additional clarity in this area.

Background

For the background and key definitions, please see the TPG's previous comments to the docket, including comments previously submitted related to minimal manipulation and homologous use.

As part of its recent draft guidance document titled *Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products: Draft Guidance for Industry and FDA Staff*, the FDA noted that "[a] homologous use for a cellular or nonstructural tissue would generally be a metabolic or biochemical function in the recipient, such as, hematopoietic, immune, and endocrine functions".¹ In general, while there may be tissue products which present such homologous uses, for allograft HCT/Ps, it is difficult to retain the 361 HCT/P designation due to the fourth key "kick up" factor (namely, "[t]he HCT/P does not have a systemic effect and is not dependent upon the metabolic activity of living cells for its primary function"). Especially for allograft products containing only cells, given the aforementioned homologous uses, it is difficult to delineate cellular only tissue products which would not be dependent upon the metabolic activity of such cells. Due to this limitation, this draft guidance document recommendation does not directly address the topic.

¹<http://www.fda.gov/downloads/BiologicsBloodVaccines/GuidanceComplianceRegulatoryInformation/Guidances/Tissue/UCM469751.pdf>

Questions and Answers

Biocompatibility

(1) Should all HCT/Ps undergo biocompatibility testing?

Currently, the FDA does not require biocompatibility testing for 361 HCT/Ps; the FDA does require biocompatibility testing for 351 HCT/Ps, consistent with ISO 10993. However, given that ISO 10993 is for devices, and not 361 HCT/Ps, there are some limitations to its application, including a broader focus (detailing interactions with metals and other materials which are not relevant to 361 HCT/Ps), lack of incorporation of innate variability in the starting material, etc. In addition, when testing in animal models, some tests may show a xenograft response, which may not be a true issue of biocompatibility. For these reasons and others, not all of the biocompatibility tests outlined in ISO 10993 will be relevant to 361 HCT/Ps.

A processor of a 361 HCT/P should perform a risk analysis, consistent with ISO 10993-1:2009 or current version, for each HCT/P or HCT/P family, to ascertain what, if any, biocompatibility testing may be appropriate. The risk analysis should focus on each HCT/P or HCT/P family, given that biocompatibility will relate to each HCT/P or HCT/P family due to different processing steps, different material incorporation rates (i.e. duration of contact, and in some cases, location of implantation), etc. In addition, a processor of a HCT/P should ensure that material intended for use as containers or packaging for HCT/Ps be compliant with USP (U.S. Pharmacopoeia) Class VI testing requirements.

Beyond the risk analysis, a processor may also want to perform biocompatibility tests (as further outlined in ISO 10993), as deemed appropriate by the risk analysis. Any biocompatibility tests should target the highest risk areas highlighted in the risk analysis.

The risk analysis will highlight which of the available biocompatibility tests may be most relevant for the HCT/P or HCT/P family. To determine which biocompatibility tests would be most relevant, a processor may want to examine issues related to material duration of contact, frequency of application, and location of implantation. Knowing the processing components that could or are encountered by the HCT/P and any existing information (publically available or internally developed) that supports history of safe use of these components can also aid in determining necessary tests. Any local and/or systemic effects of the tissue form in its final configuration should be considered. Whenever possible, testing should be performed to demonstrate the safety of any residual processing chemicals or materials. A justification as to why no further testing or no testing is required should be documented. For the format and content of the analysis, a tissue processor may want to refer to FDA's guidance document on this topic (and, in particular, Annex C of "Use of International Standard ISO 10993-1, Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process").

(2) When should a processor perform an initial, or update an existing, risk analysis?

An initial or updated risk analysis should be performed when implementing a new tissue form, instituting a change to an existing tissue process, sterilization method (if applicable), instituting a change in the physical configuration (e.g., size, geometry, surface properties), changing the manufacturer's objective intent.

Changes in tissue processes may include new chemicals, materials, or steps; changes in concentration; or changes in location of use, contact time, and temperature that can affect or alter the biocompatibility of the tissue.

(3) What is the relationship between the amount of manipulation of HCT/Ps and the need for biocompatibility testing?

An appropriate risk analysis should define the area of focus for any biocompatibility testing. In general, the extent of manipulation, whether by physical or chemical means, may increase the need for biocompatibility testing. However, for some HCT/Ps, some manipulation, such as cell removal, may improve the biocompatibility and, thus, reduce the need for additional biocompatibility testing.

(4) Are there other relevant guidance documents to consider when a tissue processor is assessing biocompatibility?

Yes. The FDA recently released a final guidance document titled “Guidance for Industry on Use of International Standard ISO 10993 1, Biological evaluation of medical devices Part 1 Evaluation and testing within a risk management process.”² This guidance covers key issues including sensitization, hemocompatibility, pyrogenicity, genotoxicity, and carcinogenicity. However, given that this guidance is specific to devices and covers a whole host of products, not all of the recommendations will be wholly relevant to HCT/Ps. For instance, for HCT/Ps, sensitization tests generally involve the use of animal models. And, as such, there may be false positive test results.

Appropriate Analysis, Characterization and Assessment

(5) What are some examples of the original, relevant characteristics of structural tissue?

Examples of the original, relevant characteristics of structural tissues include strength, flexibility, cushioning, covering, compressibility, and response to friction and shear.

(6) What about endogenous cells in structural tissues?

Structural tissues may contain endogenous cells. These endogenous cells, whether viable or non-viable, may serve to maintain the basic function of the structural tissue. Examples of these include cartilage, bone, skin for burns and amnionic membrane that contain viable or non-viable cells.

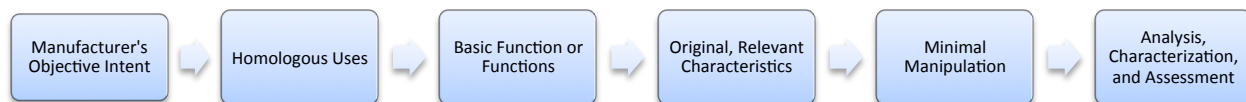
(7) How does the homologous use relate to minimal manipulation?

As noted in the flow diagram below (Diagram A), the manufacturer’s objective intent (including the associated claims) delineate the particular homologous uses of the HCT/P. Those homologous uses describe the basic function or functions relevant to such HCT/P. To ascertain whether a HCT/P is minimally manipulated, the processor must ensure that the processing steps do not alter the original, relevant characteristics that support the basic function or functions of such HCT/P. To ascertain whether those original, relevant characteristics have been retained, specific analysis, characterization, or assessment should occur and the results must be within appropriate parameters defined for success. The

²<https://www.regulations.gov/document?D=FDA-2013-D-0350-0038>

manufacturer's objective intent defines the homologous use of a HCT/P, which then delineates what appropriate processes may be considered minimal manipulation.

Diagram A. From the Manufacturer's Objective Intent to the Analysis, Characterization, and Assessment



To better understand that interaction, here are some key examples:

- Example 1-A – Acellular dermal matrix (ADM) is advertised as a sheath or cover for for supplemental support, protection, reinforcement, or covering for a tendon (a homologous use, given that a basic function of dermis is to provide strength or elasticity). The ADM undergoes certain processing steps (including decellularization). To ensure that the original, relevant characteristics are retained, a processor may perform one or more of the following tests: uniaxial tensile testing, burst strength testing, or suture retention strength testing. Provided the results are within appropriate parameters, the ADM would be considered minimally manipulated.
- Example 1-B – As part of its advertising, a processor notes that its ground ADM is used as a facial filler (a homologous use). The ADM undergoes reduction in size to ensure that the particulate size is appropriate for such use. To ensure the original, relevant characteristics are retained in the particularized ADM, the processor performs one or more of the following tests: *in vitro* proliferation and migration assays, or relevant histological characterization studies. Provided that the results are within appropriate parameters, then the ADM would be considered minimally manipulated.
- Example 1-C – The ADM is advertised as a tendon replacement (non-homologous use). The use is considered non-homologous because the required parameters for tendon replacement (e.g., the elasticity or tensile test) were not sufficient to warrant a claim related to tendon replacement.
- Example 2-A – The manufacturer's objective intent for decellularized adipose is for cushioning. The use is homologous because a basic function of adipose it to cushion. Processing steps include decellularization. To ensure the original, relevant characteristics are retained during the decellularization process, the processor performs a confined compression test, and such results are within appropriate parameters. Thus, the decellularized adipose is minimally manipulated.
- Example 2-B – The manufacturer's objective intent for decellularized adipose is "as a protective layer of tissue protection" as a dura patch. This is a non-homologous use because while adipose provides a mechanism of protection, this is significantly different from that required by the clinical application. The required function as a dura patch is to provide protection by regulating transport of fluids between two environments or augmenting a primary closure -- functions not provided by the adipose tissue.
- Example 3-A – According to the manufacturer's objective intent, a fresh osteochondral graft comprised of both bone and cartilage is used to fill or repair cartilage and osseous defects created due to injury or disease – a homologous use, given that a basic function of the graft is to facilitate healing of bone and

surrounding tissue. Very little processing occurs with such grafts. And, as such, they are minimally manipulated.

- Example 3-B – According to the manufacturer’s objective intent, a bone graft is used for anterior/posterior lumbar interbody fusion (homologous use), given that a basic function of bone is to act as a structural graft filling a bony void and facilitate bone growth by being osteoconductive. Processing steps include shaping of the bone graft, which does not alter the microstructure of the extracellular matrix. The processor performs biomechanical integrity/strength analysis, and such results are within appropriate parameters for supporting physiological loads expected in the clinical application. And, as such, the bone graft is considered minimally manipulated.
- Example 3-C – According to the package insert, demineralized bone matrix (DBM) is used in bony voids or gaps in the skeletal system (homologous use), given that a basic function of demineralized bone is to facilitate the healing of the bone by having osteoinductive potential, and by being osteoconductive, allowing revascularization and recipient cell infiltration in and around the bone graft microstructure. Processing steps include grinding and soaking the resultant bone in acid. To confirm the retention of the original, relevant characteristics, a processor performs a residual calcium test and an Athymic rodent assay or analogous *in vitro* tests (such as alkaline phosphatase or bone morphogenetic protein or BMP assays) to measure osteoinductive potential, and such results are within appropriate parameters. And, as such, the DBM is considered minimally manipulated.
- Example 3-D – According to the manufacturer’s objective intent, DBM is used for cartilage repair (non-homologous use), given that the basic function of bone is to facilitate the healing of bone and surrounding tissue, not solely cartilage.

(8) What analysis, characterization or assessment should a tissue processor perform for a particular tissue type?

The manufacturer’s objective intent defines the relevant analysis, characterization or assessment that may be appropriate for a particular tissue product. See Appendix A for some potential options. The table in Appendix A identifies potential options for analysis, characterization or assessment.³ In reviewing the recommended analysis, characterization or assessment, a processor may opt to choose only one of the recommended options to provide relevant information to support the manufacturer’s objective intent.

The examples of analyses, characterizations or assessments and the tissue forms are not all inclusive. For the analysis, characterization or assessment, a processor may also use pre-clinical testing, literature reviews, or other relevant information to support the processor’s objective intent.

Where possible, the table identifies standards or testing kits, which may be available to aid in the analysis, characterization or assessment. Other standards, processes, or procedures (including validated in-house methods) in lieu of those listed specifically in this table may also be considered, as determined by the processor.

³A manufacturer’s objective intent is determined by the expressions of the manufacturer or its representatives, or may be shown by the circumstances surrounding the distribution of the article. A manufacturer’s objective intent may, for example, be shown by labeling claims, advertising matter, or oral or written statements by the manufacturer or its representatives.

Further, the processor would only be responsible for reviewing the recommended analysis, characterization or assessment for the particular homologous uses identified by the manufacturer's objective intent, not all of the potential homologous uses identified below.

Finally, for some types of HCT/Ps that have a long-standing history of clinical use (e.g., bone and other musculoskeletal tissue used in traditional bone or soft tissue grafting procedures), such history in itself may adequately support the manufacturer's objective intent for use, such that a simple rationale in lieu of testing, particularly mechanical testing, may be most appropriate in these situations.

(9) How does the tissue processor determine the appropriate parameters for the particular analysis, characterization or assessment?

The manufacturer's objective intent as well as processing steps and methods should be taken into account when defining appropriate analysis, characterization or assessment. Additionally, the analysis, characterization or assessment should be validated or verified (with appropriate controls).

(10) When may animal studies be appropriate?

In cases where no well-established *in vitro* models exist to evaluate healing response and restoration of function, appropriate pre-clinical animal models may be considered. In addition, the FDA recently released a final guidance document titled "Guidance for Industry on Use of International Standard ISO 10993 1, Biological evaluation of medical devices Part 1 Evaluation and testing within a risk management process."⁴ This guidance covers key issues including the use of animal models (especially as it relates to degradation). However, given that this guidance is specific to devices and covers a whole host of products, not all of the recommendations will be wholly relevant to HCT/Ps. For instance, for HCT/Ps, degradation is generally not a key consideration, given that HCT/Ps do not degrade like synthetics.

(11) What is the required timing of the particular analysis, characterization or assessment? For instance, can there be subsequent processing steps after the particular analysis, characterization or assessment?

Results of any analysis, characterization or assessment should be representative of the final product.

(12) To ensure compliance with the FDA regulations related to homologous use and minimal manipulation, are there key practices that the TPG recommends all tissue processors perform?

Yes. The AATB TPG recommends that all processors develop a regulatory pathway document to delineate the manufacturer's objective intent, homologous use, original relevant characteristics, and required analysis, characterization or assessment (along with appropriate parameters) for a new or existing product derived from human tissue. In addition, this document should provide the product description, current regulatory framework (including a discussion of the four main "kick up" factors), and product labeling and promotion. This type of document provides justification detailing the regulatory approaches and processes that are reasonable, scientifically based, and meet the current FDA regulatory criteria. Finally, this type of document may also be used as a reference for review of product specific labeling and promotional materials to ensure the manufacturer's objective intent and claims fall within the current regulatory framework. The regulatory pathway document would not be submitted to the FDA but could be

⁴<https://www.regulations.gov/document?D=FDA-2013-D-0350-0038>

subject to an audit by the agency. Please see Appendix B for flow diagrams better delineating the various “kick up” factors.

Appendix A: Analysis, Characterization or Assessment Table

This table identifies potential options for analysis, characterization or assessment.⁵ In reviewing the recommended analysis, characterization, or assessment, a processor may opt to choose only one of the options to provide relevant information to support the manufacturer’s objective intent.

The examples of analysis, characterization or assessment and the tissue forms are not all inclusive. For the analysis, characterization or assessment, a processor may also use pre-clinical testing, literature reviews, or other relevant information to support the manufacturer’s objective intent.

Where possible, the table identifies standards or testing kits, which may be available to aid in the analysis, characterization or assessment. Other standards, processes, or procedures (including validated in-house methods) in lieu of those listed specifically in this table may also be considered, as determined by the processor.

Further, the processor would only be responsible for reviewing the recommended analysis, characterization or assessment for the particular homologous uses identified by the manufacturer’s objective intent, not all of the potential homologous uses identified below.

For some types of HCT/Ps that have a long-standing history of clinical use (e.g., bone and other musculoskeletal tissue used in traditional bone or soft tissue grafting procedures), such history in itself may adequately support the manufacturer’s objective intent for use, such that a simple rationale in lieu of testing, particularly mechanical testing, may be most appropriate in these situations.

Finally, for the references related to the basic function and homologous use, please refer to the *AATB TPG’s Guidance Document Recommendation: Homologous Use of Human Cells, Tissues, and Cellular and Tissue-Based Products* (October 2015). The chart below only includes additional, supportive information not already included in that previous guidance recommendation.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
Adipose	Cushioning and/or padding of tissues and/or organs	Placement for tissue cushioning and/or protection	Adipocytes containing lipids ^{6,7} Extracellular matrix ⁸	• Confined compression (ASTM F2267-04(2011)) • Indentation testing (ASTM D2240) • Relevant matrix protein characterization tests for example lipid content,

⁵A manufacturer’s objective intent is determined by the expressions of the manufacturer or its representatives, or may be shown by the circumstances surrounding the distribution of the article. A manufacturer’s objective intent may, for example, be shown by labeling claims, advertising matter, or oral or written statements by the manufacturer or its representatives.

⁶Johnson, P.R. and Greenwood, MRC. The adipose tissue. In: “Cell and Tissue Biology: A Textbook of Histology.” (ed. by L. Weiss). 6th edition, Urban and Schwarzenberg, Baltimore, MD. pp. 191-209, 1988.

⁷Symond, Michael E., ed. Adipose Tissue Biology. New York: Springer, 2012.

⁸Alkhouli N, Mansfield J, Green E, Bell J, Knight B, Liversedge N, Tham JC, Welbourn R, Shore AC, Kos K, Winlove CP. The mechanical properties of human adipose tissues and their relationships to the structure and composition of the extracellular matrix. Am. J Physiol Endocrinol Metab., 2013 Dec; 305(12):E1427-35.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
				proteoglycans, collagens type I, II, IV and VI (commercially available test kits) Relevant histological characterization
	Creates a natural aesthetic contour	Restoration of natural contours for tissue defects (e.g. face, breast, hand)	Viscosity ^{9,10} Adipocytes ^{11,12} Extracellular matrix ¹³	- Rheometry/Viscometry (ASTM D2196, D2364) - Volume retention in an animal model - Indentation testing (ASTM D2240) - Relevant matrix protein characterization tests for example lipid content, proteoglycans, collagens type I, II, IV (commercially available test kits)
	Forms a protective layer	Tissue or graft covering; scaffold to support tissue remodeling	Adipocytes ^{14,15} Extracellular Matrix ¹⁶	- Relevant histological characterization - Relevant matrix protein characterization tests for example lipid content, proteoglycans, collagens type I, II, IV (commercially available test kits)
	Allows free movement of tissue layers and/or structures	Barrier to adhesion following surgery (especially for old burn scars, release of adherent scarring, and modulation of scarring in primary cleft lip repair)	Friction coefficient ¹⁷	- Kinetic friction coefficient measurement (ASTM D1894) - Relevant histological characterization
	Contains cells that produce and	Adipose regeneration	Adipocytes ^{18,19}	Biochemical assay (commercially available test kits)

⁹Kruglikov, I.L. (2014) General Theory of Body Contouring: 2. Modulation of mechanical properties of subcutaneous fat tissue. Journal of Cosmetics, Dermatological Sciences and Applications, 4, 117-127. <http://dx.doi.org/10.4236/jcdsa.2014.42017>

¹⁰Sommer G, Eder M, Kovacs L, Pathak H, Bonitz L, Mueller C, Reginig P, Holzapfel, GA. Multiaxial mechanical properties and constitutive modeling of human adipose tissue: A basis for preoperative simulations in plastic and reconstructive surgery. Acta Biomaterialia, 2013 Nov; 9 (11):9036-9048.

¹¹Johnson, P.R. and Greenwood, MRC. The adipose tissue. In: "Cell and Tissue Biology: A Textbook of Histology." (ed. by L. Weiss). 6th edition, Urban and Schwarzenberg, Baltimore, MD. pp. 191-209, 1988.

¹²Symond, Michael E., ed. Adipose Tissue Biology. New York: Springer, 2012.

¹³Alkhouli N, Mansfield J, Green E, Bell J, Knight B, Liversedge N, Tham JC, Welbourn R, Shore AC, Kos K, Winlove CP. The mechanical properties of human adipose tissues and their relationships to the structure and composition of the extracellular matrix. Am. J Physiol Endocrinol Metab., 2013 Dec; 305(12):E1427-35.

¹⁴Johnson, P.R. and Greenwood, MRC. The adipose tissue. In: "Cell and Tissue Biology: A Textbook of Histology." (ed. by L. Weiss). 6th edition, Urban and Schwarzenberg, Baltimore, MD. pp. 191-209, 1988.

¹⁵Symond, Michael E., ed. Adipose Tissue Biology. New York: Springer, 2012.

¹⁶Alkhouli N, Mansfield J, Green E, Bell J, Knight B, Liversedge N, Tham JC, Welbourn R, Shore AC, Kos K, Winlove CP. The mechanical properties of human adipose tissues and their relationships to the structure and composition of the extracellular matrix. Am. J Physiol Endocrinol Metab., 2013 Dec; 305(12):E1427-35.

¹⁷Kochhar A, Wu I, Mohan R, Conde-Green A, Hillel AT, Byrne PJ, Elisseeff JH. A comparison of the rheologic properties of an adipose-derived extracellular matrix biomaterial lipoaspirate, calcium hydroxylapatite, and cross-linked hyaluronic acid. JAMA Facial Plast Surg. 2014 Nov-Dec;16(6):405-9.

¹⁸Johnson, P.R. and Greenwood, MRC. The adipose tissue. In: "Cell and Tissue Biology: A Textbook of Histology." (ed. by L. Weiss). 6th edition, Urban and Schwarzenberg, Baltimore, MD. pp. 191-209, 1988.

¹⁹Symond, Michael E., ed. Adipose Tissue Biology. New York: Springer, 2012.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
	maintain its own extracellular matrix		Cells of the Stromal Vascular Fraction (SVF) ²⁰	- Cell viability assay (LDH, ATP assay)(commercially available test kits) - Lipolysis^{21,22} - Flow cytometry to characterize cell types (commercially available test kits) - Relevant histological and immunohistological characterization.
	Contains cells that sustain all of its basic functions and allow for self-renewal	All of the above	Adipocytes ^{23,24} Cells of the Stromal Vascular Fraction (SVF) ²⁵	- Biochemical assay (commercially available test kits) - Cell viability assay (LDH, ATP assay)(commercially available test kits) - Lipolysis^{26,27} - Flow cytometry to characterize cell types (commercially available test kits) - Colony Forming Unit (CFU) in cell culture to test the ability of SVF cells to proliferate - Relevant histological and immunohistological characterization
Amnionic or placental membrane	Protective barrier	Tissue or graft covering; scaffold to support tissue remodeling	Porosity, osmotic permeability, Material stability ^{28,29,30}	- Porosity characterization (ASTM F2450) - Relevant histological characterization - Osmosis permeability chamber assay (commercially available test kit)

²⁰Bourin, P., Bunnell, B.A., Casteilla, L., Dominici, M., Katz, A.J., March, K.L., Redl, H., Rubin, P.j., Yoshimura, K., Gimble, J.M. Stromal cells from the adipose tissue-derived stromal vascular fraction and culture expanded adipose tissue-derived stromal/stem cells: a joint statement of the International Federation for Adipose Therapeutics (IFATS) and Science and the International Society for Cellular Therapy (ISCT). *Cytotherapy*, 2013 Jun; 15(6): 641-648

²¹Schweiger M, Eichmann TO, Taschler U, Zimmermann R, Zechner R, Lass A. Measurement of Lipolysis. *Methods in Enzymology*. 2014; 538:171-93. doi:10.1016/B978-0-12-800280-3.00010-4.

²²Zhu M, Cohen SR, Hicok KC. Comparison of three fat graft preparation methods: gravity separation, centrifugation, and simultaneous washing with filtration in a closed system. *Plast Reconstr Surg*. 2013 Apr; 131(4): 873-80.

²³Johnson, P.R. and Greenwood, MRC. The adipose tissue. In: "Cell and Tissue Biology: A Textbook of Histology." (ed. by L. Weiss). 6th edition, Urban and Schwarzenberg, Baltimore, MD. pp. 191-209, 1988.

²⁴Symond, Michael E., ed. *Adipose Tissue Biology*. New York: Springer, 2012.

²⁵Bourin, P., Bunnell, B.A., Casteilla, L., Dominici, M., Katz, A.J., March, K.L., Redl, H., Rubin, P.j., Yoshimura, K., Gimble, J.M. Stromal cells from the adipose tissue-derived stromal vascular fraction and culture expanded adipose tissue-derived stromal/stem cells: a joint statement of the International Federation for Adipose Therapeutics (IFATS) and Science and the International Society for Cellular Therapy (ISCT). *Cytotherapy*, 2013 Jun; 15(6): 641-648

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³⁰Cell Tissue Bank. 2003;4(2-4):95-100. Properties of Air Dried Radiation Processed Amniotic Membranes under Different Storage Conditions. Singh R1, Gupta P, Kumar P, Kumar A, Chacharkar MP.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
	Allows free movement	Reduced risk of adhesions following surgery	Low coefficient of friction ^{31,32} Mechanical barrier ^{33,34,35}	- Kinetic friction coefficient measurement (ASTM D1894) - Relevant histological characterization - Porosity characterization (ASTM F2450) - Osmosis permeability chamber assay (commercially available test kit)
	Prevents infection	Reduced risk of infection	Mechanical barrier ^{36,37,38} Antimicrobial components ^{39,40,41,42}	- Porosity characterization (ASTM F2450) - Osmosis permeability chamber assay (commercially available test kit) - Antimicrobial efficacy testing (ZOI, AATCC Method 100, ASTM 2180, USP <51>)
	Rapid self-restoration	Healing of wounds or burns with minimal scarring and inflammation; ocular surface reconstruction	Rapid self-restoration ^{43,44} Active growth factors, cytokines, chemokines and other factors known to modulate the inflammation and healing process ^{45,46,47}	<i>In vivo</i> model for wounds with histological evaluation (with endpoints of epithelialization, wound closure rate, histological evaluation of healed tissue architecture) <i>In vitro</i> measurements of modulation

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³⁵Cell Tissue Bank. 2003;4(2-4):95-100. Properties of Air Dried Radiation Processed Amniotic Membranes under Different Storage Conditions. Singh R1, Gupta P, Kumar P, Kumar A, Chacharkar MP.

³⁶Minutes, Meeting of the Cellular, Tissue and Gene Therapies Advisory Committee, CBER, FDA, May 14, 2009;p 126.

³⁷J Tissue Viability. 2011 May;20(2):49-54. doi: 10.1016/j.jtv.2010.06.001. Epub 2010 Jul 8. Dried gamma-irradiated amniotic membrane as dressing in burn wound care. Singh R1, Chacharkar MP.

³⁸Cell Tissue Bank. 2003;4(2-4):95-100. Properties of Air Dried Radiation Processed Amniotic Membranes under Different Storage Conditions. Singh R1, Gupta P, Kumar P, Kumar A, Chacharkar MP.

³⁹Eur Cell Mater. 2008 Apr 29;15:88-99. Properties of the amniotic membrane for potential use in tissue engineering. Niknejad H1, Peirovi H, Jorjani M, Ahmadiani A, Ghanavi J, Seifalian AM.

⁴⁰J Biomed Mater Res B Appl Biomater. 2014 Aug;102(6):1353-62. doi: 10.1002/jbm.b.33141. Epub 2014 Mar 25.

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⁴¹Vasc Cell. 2014 May 1;6:10. doi: 10.1186/2045-824X-6-10. eCollection 2014. Angiogenic properties of dehydrated human amnion/chorion allografts: therapeutic potential for soft tissue repair and regeneration. Koob TJ1, Lim JJ1, Massee M1, Zabek N1, Rennert R2, Gurtner G2, Li WW3.

⁴²Int Wound J. 2013 Oct;10(5):493-500. doi: 10.1111/iwj.12140. Epub 2013 Aug 1. Biological properties of dehydrated human amnion/chorion composite graft: implications for chronic wound healing. Koob TJ1, Rennert R, Zabek N, Massee M, Lim JJ, Temenoff JS, Li WW, Gurtner G.

⁴³Eur Cell Mater. 2008 Apr 29;15:88-99. Properties of the amniotic membrane for potential use in tissue engineering. Niknejad H1, Peirovi H, Jorjani M, Ahmadiani A, Ghanavi J, Seifalian AM.

⁴⁴November 26, 2001 FDA Letter to Bio-Tissue, Inc.

⁴⁵J Biomed Mater Res B Appl Biomater. 2014 Aug;102(6):1353-62. doi: 10.1002/jbm.b.33141. Epub 2014 Mar 25.

Properties of dehydrated human amnion/chorion composite grafts: Implications for wound repair and soft tissue regeneration. Koob TJ1, Lim JJ, Massee M, Zabek N, Denozière G.

⁴⁶Vasc Cell. 2014 May 1;6:10. doi: 10.1186/2045-824X-6-10. eCollection 2014.

Angiogenic properties of dehydrated human amnion/chorion allografts: therapeutic potential for soft tissue repair and regeneration.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
				factors <i>In vitro</i> proliferation and migration assays
	Contains cells that sustain all of its basic functions	All of the above	Amniotic Epithelial Cells, Mesenchymal Stromal Cells ⁴⁸	- Relevant histological characterization - Cell viability assay (commercially available test kits) - Flow cytometry to characterize cell types (commercially available test kits)
Articular cartilage ⁴⁹	Presents smooth surface	Cartilage restoration by osteochondral or shell grafting	Low coefficient of friction ^{50,51} Surface profile and smoothness ^{52,53}	- Kinetic friction coefficient measurement (ASTM 1894) - Surface roughness measurement (ASTM F2791-15)
	Cushioning	Osteochondral defect filling ^{54,55,56,57}	Compression strength; Proteoglycan and glycosaminoglycan content ⁵⁸	- Confined compression testing (ASTM F2267-04(2011)) - Indentation testing (ASTM D2240) - GAG and HA quantification biochemical assay (commercially available test kits) - Safranin O staining for proteoglycans responsible for cushioning effect (commercially available test kits) - Evaluations post in vivo testing including but not limited to those listed in ASTM F2451

Koob TJ1, Lim JJ1, Massee M1, Zabek N1, Rennert R2, Gurtner G2, Li WW3.

⁴⁷Int Wound J. 2013 Oct;10(5):493-500. doi: 10.1111/iwj.12140. Epub 2013 Aug 1. Biological properties of dehydrated human amnion/chorion composite graft: implications for chronic wound healing. Koob TJ1, Rennert R, Zabek N, Massee M, Lim JJ, Temenoff JS, Li WW, Gurtner G.

⁴⁸Bourne, G., The foetal membranes. A review of the anatomy of normal amnion and chorion and some aspects of their function. Postgrad Med J, 1962. 38: p. 193-201.

⁴⁹Response to friction and shear is a less specific term that is relevant to this that the FDA included in their examples of original relevant characteristics.

⁵⁰Wang, H., & Ateshian, G. A. (1997). The normal stress effect and equilibrium friction coefficient of articular cartilage under steady frictional shear. Journal of biomechanics, 30(8), 771-776.

⁵¹<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3445147/>

⁵²Brill, N., Riedel, J., Schmitt, R., Tingart, M., Truhn, D., Pufe, T., ... & Nebelung, S. (2015). 3D Human cartilage surface characterization by optical coherence tomography. Physics in medicine and biology, 60(19), 7747.

⁵³Chu CR, Szczodry M, Bruno S. Animal Models for Cartilage Regeneration and Repair. Tissue Engineering Part B, Reviews. 2010;16(1):105-115. doi:10.1089/ten.teb.2009.0452.

⁵⁴Riboh, Jonathan C., Brian J. Cole, and Jack Farr. "Particulated articular cartilage for symptomatic chondral defects of the knee." Current reviews in musculoskeletal medicine 8.4 (2015): 429-435.

⁵⁵Stroh, D. Alex, Aaron J. Johnson, and Michael A. Mont. "Surgical implants and technologies for cartilage repair and preservation of the knee." Expert review of medical devices 8.3 (2011): 339-356.

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⁵⁷Williams RJ III, Ranawat AS, Potter HG, Carter T, Warren RF. Fresh stored allografts for the treatment of osteochondral defects of the knee. J. Bone Joint Surg. Am.89(4), 718–726 (2007).

⁵⁸Williams, S. K., Amiel, D., Ball, S. T., Allen, R. T., Wong, V. W., Chen, A. C., ... & Bugbee, W. D. (2003). Prolonged storage effects on the articular cartilage of fresh human osteochondral allografts. J Bone Joint Surg Am,85(11), 2111-2120.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
	Contains cells that produce and maintain its own extracellular matrix	Cartilage regeneration	Contains chondrocytes ⁵⁹	- Relevant histological characterization or biochemical assay (commercially available test kits) - Cell viability assay (LDH, ATP assay)(commercially available test kits) - Flow cytometry to characterize cell types (commercially available test kits)
Blood vessels	Act as a valved or non-valved conduits.	Femoral veins can act as replacement pulmonary heart valves (PHV) to replace congenitally malformed PHVs. Arteries and veins can act as dialysis access grafts, peripheral bypass grafts, valved or non-valved extracardiac shunts, coronary artery by-pass grafts and as conduits for blood and other bodily fluids, in general.	Biomechanical properties identified in ANSI/AAMI/ISO 7198 and 5840 ⁶⁰	- Biomechanical testing (ISO 7198 and 5840) as relevant the manufacturer's objective intent - Differential scanning calorimetry (examples listed in ASTM F2150)
Bone	Facilitates healing of bone and surrounding tissue	- Joint reconstruction - Closure of oral/nasal fistulas - Non-load bearing filler for boney tissues	Osteoconductivity ⁶¹	- Biomechanical integrity/strength analysis - Histology, microCT, or X Ray (ASTM F2450-10)

⁵⁹Buckwalter, J. A., & Mankin, H. J. (1997). Articular cartilage: tissue design and chondrocyte-matrix interactions. Instructional course lectures, 47, 477-486.

⁶⁰http://www.ehow.com/about_5082107_function-blood-vessels.html

⁶¹Prendergast, Mechanics of Bone Regeneration, Ch. 32 and other sections, in Bone Mechanics Handbook, Second Edition, Stephen C. Cowin, ed., CRC Press 2001.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
		- Treatment of bone cysts with DBM - DBM use in extremities, pelvis, and long bone repair - Scoliosis correction (with DBM) - Anthroplasty applications (with DBM) - DBM use in sports medicine	Mineral content ⁶² Osteoinductive potential Growth factor content (e.g., BMP-2, -4, -7, TGF-beta, etc.) ^{63,64,65,66,67,68,69,70,71,72}	- Athymic rodent assay (ASTM F2529-13) - Residual calcium (commercially available test kit; ASTM WK21829 for atomic absorption) - BMP-2 (commercially available test kit) - Alkaline Phosphase (ALP) (commercially available test kit)
			Graft Density ^{73,74,75,76,77}	Mass and volume recording
	Structural support for the skeletal system and protection for internal organs	- Repair of trauma in long bones - Anterior/posterior lumbar interbody fusion	Mechanical properties/bone ⁷⁸	- Compression testing (ASTM D695 or D1621) - Insertion testing (ASTM F2777-10) - Torsion testing (ASTM F2942-13) - Flexural/Bending testing (examples listed within ASTM F2150)
	Contains endogenous cells	- Joint fusion - Spinal fusion	Cell characterization (including mesenchymal stem cells,	Relevant histological characterization or biochemical assay (commercially

⁶²Boskey, Bone Mineralization, Ch. 5, Guo, Mechanical Properties of Cortical Bone and Cancellous Bone Tissue, Ch. 10, and other sections in Bone Mechanics Handbook, Second Edition, Stephen C. Cowin, ed., CRC Press 2001

⁶³Prendergast, Mechanics of Bone Regeneration, Ch. 32 and other sections, in Bone Mechanics Handbook, Second Edition, Stephen C. Cowin, ed., CRC Press 2001.

⁶⁴Chnari, Bone Morphogenetic Protein 2 (BMP-2) Levels are Predictive of the Osteoinductive Potential of Demineralized Bone Matrix Poster No . 485 • 56th Annual Meeting of the Orthopaedic Research Society.

⁶⁵Han, Quantitative and sensitive in vitro assay for osteoinductive activity of demineralized bone matrix, J Orthop res, 21: 648-654, 2003.

⁶⁶Han, Intervariability and intravariability of bone morphogenetic proteins in commercially available demineralized bone matrix products, Spine, 31: 1299-1306, 2006.

⁶⁷Bae, Variability across ten production lots of a single demineralized bone matrix product, JBJS, 92: 427-435, 2010.

⁶⁸Katz, Demineralized bone matrix as an osteoinductive biomaterial and in vitro predictors of its biological potential. JBMR 89: 127-134, 2009.

⁶⁹Peterson, Osteoinductivity of commercially available demineralized bone matrix, JBJS, 86: 2243-2250, 2004.

⁷⁰Pietrzak, Assay of Bone Morphogenetic Protein-2, -4, and -7 in Human Demineralized Bone Matrix, J Craniofac Surg, 17: 84-90, 2006

⁷¹Wildemann, Quantification of various growth factors in different demineralized bone matrix preparations, JBMR, 81A: 437-442, 2006.

⁷²Blum, Measurement of bone morphogenetic proteins and other growth factors in demineralized bone matrix, Orthopedics, 27: S161-S165, 2004.

⁷³Bone Mechanics Handbook, Second Edition, Stephen C. Cowin, ed., CRC Press 2001

⁷⁴Blanton, Density of Fresh and Embalmed Human Compact and Cancellous Bone, Am. J. Phys. Anthropol., 29: 39-44, 1968

⁷⁵Dhakar, Bone Density Changes after Radiation for Extremity SarcomaS: Exploring the Etiology of Pathologic Fractures., Int., J. Radiation Oncology Biol. Phys. 80(4): 1158-11663, 2011.,

⁷⁶Mosekilde, Sex Difference in Age-related Changes in Vertebral Body Size, Density, and Biomechanical Competence in Normal Individuals, Bone, 11: 67-73, 1990.

⁷⁷McCubrey, Static and Fatigue Failure Properties of Thoracic and Lumbar Vertebral Bodies and their relation to regional density, J. Biomech., 28(8): 891-899, 1995.

⁷⁸Guo, Mechanical Properties of Cortical Bone and Cancellous Bone Tissue, Ch. 10, and other sections in Bone Mechanics Handbook, Second Edition, Stephen C. Cowin, ed., CRC Press 2001

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
	which support self-restoration and maintenance of normal bone matrix		osteogenic cells, osteoprogenitor cells, osteoblasts, bone progenitor cells, etc.) ^{79,80,81,82,83,84,85}	- available test kits) - Cell viability assay (LDH, ATP) - Cell content assay)(commercially available test kits) - Flow cytometry to characterize cell types (commercially available test kits)
Costal cartilage or other non-articular cartilage	Supplies non-rigid support and performs shock absorption	- Interpositional spacer - Tympanoplasty - Rhinoplasty	Yield strength, flexibility ⁸⁶	- Indentation testing (ASTM D2240) - Flexural/bending testing⁸⁷ (examples listed within ASTM F2150)
	Defines anatomical cavities and shapes	Physical reconstruction of non-rigid tissues such as nose, ear, trachea and nipple)	Yield strength, flexibility ⁸⁸	- Indentation testing (ASTM D2240) - Flexural/bending testing (examples listed within ASTM F2150) - Visual inspection
	Contains cells (including chondrocytes) that produce and maintain its own extracellular matrix	Cartilage regeneration	Contains chondrocytes ⁸⁹	- Relevant histological characterization or biochemical assay (commercially available test kits) - Cell viability assay (LDH, ATP assay)(commercially available test kits) - Flow cytometry to characterize cell types (commercially available test kits)
			Contains proteoglycans, collagen type II ⁹⁰	GAG and HA quantification biochemical assay (commercially available test kits)

⁷⁹Barry, F.P. and J.M. Murphy, Mesenchymal stem cells: clinical applications and biological characterization. Int J Biochem Cell Biol, 2004. 36(4): p. 568-84

⁸⁰Jaiswal, N., et al., Osteogenic differentiation of purified, culture-expanded human mesenchymal stem cells in vitro. J Cell Biochem, 1997. 64(2): p. 295-312.

⁸¹Marie, P.J. and O. Fromigue, Osteogenic differentiation of human marrow-derived mesenchymal stem cells. Regen Med, 2006. 1(4): p. 539-48

⁸²Pittenger, M.F., et al., Multilineage potential of adult human mesenchymal stem cells. Science, 1999. 284(5411): p. 143-7.

⁸³Risbud, M.V., et al., Osteogenic potential of adult human stem cells of the lumbar vertebral body and the iliac crest. Spine (Phila Pa 1976), 2006. 31(1): p. 83-9.

⁸⁴Malaval, L., et al., Kinetics of osteoprogenitor proliferation and osteoblast differentiation in vitro. J Cell Biochem, 1999. 74(4): p. 616-27

⁸⁵Claros, S., et al., Selection and induction of rat skeletal muscle-derived cells to the chondro-osteogenic lineage. Cell Mol Biol (Noisy-le-grand), 2008. 54(1): p. 1-10

⁸⁶<http://www.healthline.com/human-body-maps/costal-cartilage>

⁸⁷Williams RJ III, Ranawat AS, Potter HG, Carter T, Warren RF. Fresh stored allografts for the treatment of osteochondral defects of the knee. J. Bone Joint Surg. Am.89(4), 718–726 (2007).ASTM F2150 is a general standard for “Characterization and Testing of Biomaterial Scaffolds Used in Tissue-Engineered Medical Products” Within this standard, they list the following ASTM: Flexural/Bending Properties—Depending on a scaffold’s physical or dimensional characteristics, its flexural properties may be evaluated using methodology found in one or more of the following Test Methods: D648, D671, D747, D790, D1388, D5732, and D6125.

⁸⁸<http://www.healthline.com/human-body-maps/costal-cartilage>

⁸⁹Kitaoka, E., Satomura, K., Hayashi, E., Yamanouchi, K., Tobiume, S., Kume, K., ... & Nagayama, M. (2001). Establishment and characterization of chondrocyte cell lines from the costal cartilage of SV40 large T antigen transgenic mice. Journal of cellular biochemistry, 81(4), 571-582.

⁹⁰Rejtarová, O., Hejna, P., Soukup, T., & Kuchař, M. (2009). Age and sexually dimorphic changes in costal cartilages. A preliminary microscopic study. Forensic science international, 193(1), 72-78.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
				Safranin O staining for proteoglycans responsible for cushioning effect (commercially available test kits)
Dermis	Provides strength and elasticity	- Bladder sling - Breast reconstruction - Tissue expander to allow for enhanced fill volume - Protects abdominal contents and restores functional support - Abdominal wall re-enforcement - Provides augmentation of tendon and rotator cuff repair - Provides a filler for congenital and trauma-induced defects - Dental procedures - Provides a flap for guided tissue regeneration in the mouth - Hand reconstruction - Rotator cuff re-enforcement	Tissue Uniformity ^{91,92}	Visual inspection
			Tensile strength ^{93,94}	- Uniaxial tensile testing (ASTM D882) - Burst strength testing (ASTM D6797) - Suture retention strength testing (ISO 7198)
	Matrix to support epithelialization	Treatment of chronic wounds^{95,96,97,98,99}	Extracellular Matrix Integrity ^{100,101}	Uniaxial tensile testing (ASTM D882 or D638)

⁹¹ Brigido SA, Boc SF, Lopez RC. Effective management of major lower extremity wounds using an acellular regenerative tissue matrix: a pilot study. Orthopedics. 2004 Jan; 27(1 Suppl):s145-9. PubMed PMID: 14763548.

⁹² Jodi Walters DPM1, Shawn Cazzell, DPM2, Hau Pham, DPM3, Dean Vayser, DPM, FACFAS4, Alexander Reyzelman, DPM, FACFAS5 Healing rates in a multicenter assessment of a sterile, room temperature, acellular dermal matrix versus conventional care wound management and an active comparator in the treatment of full thickness diabetic foot ulcers. EPlasty Feb 2016.

⁹³ Sambit Sahoo, Katherine R. DeLozier, Ahmet Erdemir, Kathleen A. Derwin Department of Biomedical Engineering, Cleveland Clinic, Cleveland, OH 44195, USA

⁹⁴ Department of Biomedical Engineering, University of Akron, Akron, OH 44325, USA. Clinically relevant mechanical testing of hernia graft constructs. Elsevier Research Paper, 27 October 2015. <http://dx.doi.org/10.1016/j.jmbbrn.2014.10.011>

⁹⁵ Brigido SA. The use of an acellular dermal regenerative tissue matrix in the treatment of lower extremity wounds: a prospective 16-week pilot study. Int Wound J. 2006 Sep; 3(3):181-7. PubMed PMID: 16984575.

⁹⁶ Brigido SA, Boc SF, Lopez RC. Effective management of major lower extremity wounds using an acellular regenerative tissue matrix: a pilot study. Orthopedics. 2004 Jan; 27(1 Suppl):s145-9. PubMed PMID: 14763548.

⁹⁷ Martin BR, Sangalang M, Wu S, Armstrong DG. Outcomes of allogenic acellular matrix therapy in treatment of diabetic foot wounds: an initial experience. Int Wound J. 2005 Jun; 2(2):161-5. PubMed PMID: 16722865.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
			Porosity ¹⁰²	- Burst strength testing (ASTM D6797) - Suture retention strength testing (ISO 7198) - Porometry (ASTM E128, E1294, F316), Porosity characterization (ASTM F2450) - Relevant histological characterization - Osmosis permeability chamber assay (commercially available test kit)
	Filler ^{103,104}	Soft tissue defect filler including the extremities, throat, face or breast^{105,106}	Extracellular Matrix Integrity ^{107,108}	- Dissolution <i>In vitro</i> proliferation and migration assays - Relevant histological characterization
Fascia lata	Attach, stabilize, enclose, and separate muscles and other internal organs; reduces friction between muscles allowing	- Interpositional arthroplasty in foot, hand and shoulder joints - To provide augmentation to soft tissues where	Tensile strength ¹¹⁰	- Suture retention strength testing (ISO 7198) - Kinetic friction coefficient measurement (ASTM D1894) - Relevant histological characterization - Uniaxial Testing (ASTM D882 or D638) - Burst strength testing (ASTM D6797)

⁹⁸ Reyzelman A, Crews RT, Moore JC, Moore L, Mukker JS, Offutt S, Tallis A, Turner WB, Vayser D, Winters C, Armstrong DG. Clinical effectiveness of an acellular dermal regenerative tissue matrix compared to standard wound management in healing diabetic foot ulcers: a prospective, randomised, multicentre study. *Int Wound J*. 2009 Jun; 6(3):196-208. Epub 2009 Apr 2. PubMed PMID: 19368581

⁹⁹ Winters CL, Brigido SA, Liden BA, Simmons M, Hartman JF, Wright ML. A multicenter study involving the use of a human acellular dermal regenerative tissue matrix for the treatment of diabetic lower extremity wounds. *Adv Skin Wound Care*. 2008 Aug; 21(8):375-81. PubMed PMID: 18679086.

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¹⁰¹ Department of Biomedical Engineering, University of Akron, Akron, OH 44325, USA. Clinically relevant mechanical testing of hernia graft constructs. Elsevier Research Paper, 27 October 2015. <http://dx.doi.org/10.1016/j.jmbbrn.2014.10.011>

¹⁰² OECD. (2004), Guidance Document for the Conduct of Skin Absorption Studies, OECD Series on Testing and Assessment, No. 28, OECD Publishing, Paris. DOI: <http://dx.doi.org/10.1787/9789264078796-en>

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Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
	sliding	weakness of defect exists (defect could be anatomical/degenerative or surgically created). ¹⁰⁹	Parallel fiber orientation ¹¹¹	- Visual assessment - Microscopic assessment
Heart valves	Maintain forward flow of blood through the cardiovascular system	Replacement of diseased, damaged or congenitally malformed heart valve	Valve functionality ^{112,113}	- Biomechanical testing (ISO 5840) as relevant to the manufacturer's objective intent - Differential scanning calorimetry (examples listed in ASTM F2150)
		Use of arterial elements to reconstruct blood conduits (2D and tubular) Blood vessel augmentation, reconstruction and/or replacement	Tensile strength ^{114,115}	- Biaxial (ASTM D3787) - Suture pullout testing (ISO 5840) - Differential scanning calorimetry (examples listed in ASTM F2150)
Meniscus	Disperse the weight of the body and reduce friction during movement	- Meniscal transplant - Interpositional arthroplasty in foot, hand, and shoulder joints	Compression strength ¹¹⁶	- Compression testing (ASTM D695 or D1621) - Contact pressure measurement - Uniaxial Testing (ASTM D882 or D638)
			Contains GAG and collagen ¹¹⁷	- Glycosaminoglycan (GAG) content (commercially available test kits) - Collagen content (ASTM F2212-11 and commercially available test kits)

¹¹⁰Derwin, K. A., Baker, A. R., Spragg, R. K., Leigh, D. R., Farhat, W. and Iannotti, J. P. (2008), Regional variability, processing methods, and biophysical properties of human fascia lata extracellular matrix. *J. Biomed. Mater. Res.*, 84A: 500–507. doi: 10.1002/jbm.a.31455

¹⁰⁹This includes rotator cuff repair, repair of abdominal/diaphragmatic hernias, reconstruction of chest wall defects, reconstruction and reinforcement of hiatal defects, reconstruction of Achilles Tendon, repair and/or reinforcement of the dural membrane, and soft tissue defects of the hand.

¹¹¹Moore, KL; Dalley, AF; Agur, AM (2013). *Clinically Oriented Anatomy* (7th ed.). Lippincott Williams & Wilkins. p. 532. ISBN 978-1-4511-8447-1.

¹¹²Hinton RB, Yutzey KE. Heart Valve Structure and Function in Development and Disease. *Annual review of physiology*. 2011;73:29-46. doi:10.1146/annurev-physiol-012110-142145.

¹¹³<http://www.gwheartandvascular.org/education/anatomy-and-function-of-the-heart-valves/>

¹¹⁴Hinton RB, Yutzey KE. Heart Valve Structure and Function in Development and Disease. *Annual review of physiology*. 2011;73:29-46. doi:10.1146/annurev-physiol-012110-142145.

¹¹⁵<http://www.gwheartandvascular.org/education/anatomy-and-function-of-the-heart-valves/>

¹¹⁶<http://www.ivysportsmed.com/en/knee-pain/meniscus-function>

¹¹⁷Bursac P, Arnoczky S, York A (2009) Dynamic compressive behavior of human meniscus correlates with its extra-cellular matrix composition. *Biorheology* 46:227–237.

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
OsteoArticular or Osteochondral Tissue	Protective barrier for bone in articulating surfaces	- Fill or repair cartilage and osseous defects created due to injury or disease - Replaces bone and articular surfaces in reconstructive surgery for malignant bone tumors - Repair cartilage and osseous defects created due to injury or degenerative disease	Cell characterization, especially for chondrocytes ^{118,119,120}	- Relevant histological characterization or biochemical assay (commercially available test kits) - Indentation testing (ASTM D2240) - Cell viability assay (LDH, ATP assay)(commercially available test kits) - Flow cytometry to characterize cell types (commercially available test kits)
Pericardium	Covering ^{121,122}	- Intracardiac reinforcement - Extracardiac reinforcement	Structural integrity Thickness ^{123,124,125,126}	- Differential scanning calorimetry (examples listed in ASTM F2150) - Dimensional measurement
Split thickness skin (epidermis and dermis) ¹²⁷	Protective barrier against the external environment	- Limits microbial contamination of wounds - Burn treatment - Provides an extracellular matrix - Burn management - Treatment of	Porosity, permeability ¹²⁸ Cell characterization ¹²⁹	- Porosimetry (ASTM D2873) - Porometry (ASTM E128, E1294, F316), Porosity characterization (ASTM F2450) - Relevant histological characterization - Osmosis permeability chamber assay (commercially available test kit) - Relevant histological characterization

¹¹⁸Bright RW, Burchardt H: The Biomechanical Properties of Preserved Bone Grafts. In Friedlaender GE, Mankin HJ, Sell KW (eds). Osteochondral Allografts: Biology, Banking, and Clinical Applications. Boston: Little, Brown and Company; 1983. 241-247.

¹¹⁹Seth K. Williams, David Amiel, Scott T. Ball, R. Todd Allen, Van W. Wong, Albert C. Chen, Robert L. Sah, and William D. Bugbee. Prolonged Storage Effects on the Articular Cartilage of Fresh Human Osteochondral Allografts. J. Bone Joint Surg. Am., Nov 2003; 85: 2111 - 2120.

¹²⁰SP. The biology of allograft incorporation. J Knee Surg 2006; 19(3): 207-14.

¹²¹Clin. Cardiol. Vol. 22 (Suppl. I), 1-4-1-5 (1999). Function of the Normal Pericardium Ralph Shabetai,

¹²²Lee Jm and Boughner DR. Mechanical Properties of Human Pericardium, Differences in Viscoelastic Response When Compared with Canine Pericardium. Circulation Research. 57(3) September 1985.

¹²³Sato, M., Hiramatsu, Y., Matsushita, S., Sato, S., Watanabe, Y., & Sakakibara, Y. (2014). Shrinkage temperature and anti-calcification property of triglycidylamine-crosslinked autologous tissue. Journal of Artificial Organs, 17(3), 265-271.

¹²⁴Aguiari P, Fiorese M, Lop L, Gerosa G, and Bagno A. Mechanical testing of pericardium for manufacturing prosthetic heart valves. Interactive CardioVascular and Thoracic Surgery 22 (2016) 72-84.

¹²⁵Vinci MC, Tessitore G, Castiglioni L, Prandi F, Soncini M, Santoro R, Consolo F, Colazzo F, Micheli B, Sironi L, Polvani G, and Pesce M. Mechanical Compliance and Immunological Compatibility of Fixative-Free Decellularized/Cryopreserved Human Pericardium. PLoS ONE 8(5): e64769. Doi:10.1371/journal.pone.0064769.

¹²⁶Hasan A, Ragaert K, Swieszkowski W, Selimovic S, Paul A, Camci-Unal G, Mofrad M, Khademhosseini A. Biomechanical properties of native and tissue engineered heart valve constructs. Journal of Biomechanics 47 (2014): 1949-1963.

¹²⁷Previously "Epidermis."

¹²⁸<https://www.merckmanuals.com/home/skin-disorders/biology-of-the-skin/structure-and-function-of-the-skin>

¹²⁹<https://www.merckmanuals.com/home/skin-disorders/biology-of-the-skin/structure-and-function-of-the-skin>

Tissue Category	Basic Function	Homologous Use	Original, relevant characteristics (Structural) Relevant characteristics of the cells or tissues (Non-Structural)	Examples of Analysis, Characterization or Assessment
		chronic wounds		
Tendons or ligaments	High tensile strength while transmitting forces	• Anterior cruciate ligament repair • Medial collateral ligament repair • Posterior cruciate ligament repair • Medial patellofemoral ligament repair • Lateral collateral ligament repair • Foot and ankle repair • Elbow repair • Hand/wrist repair • Sternoclavicular joint instability • Quad tendon repair • Bicep tendon repair	Elasticity/Tensile strength ¹³⁰	• General standard (ASTM F2903) • Torsion testing (ASTM F2942-13) • Creep characterization (as listed in ASTM F2903) • Suture retention strength (ISO 7198) • Relevant histological characterization • Uniaxial tensile Testing (ASTM D882 or D638)
			Fiber orientation ^{131,132}	• Visual inspection

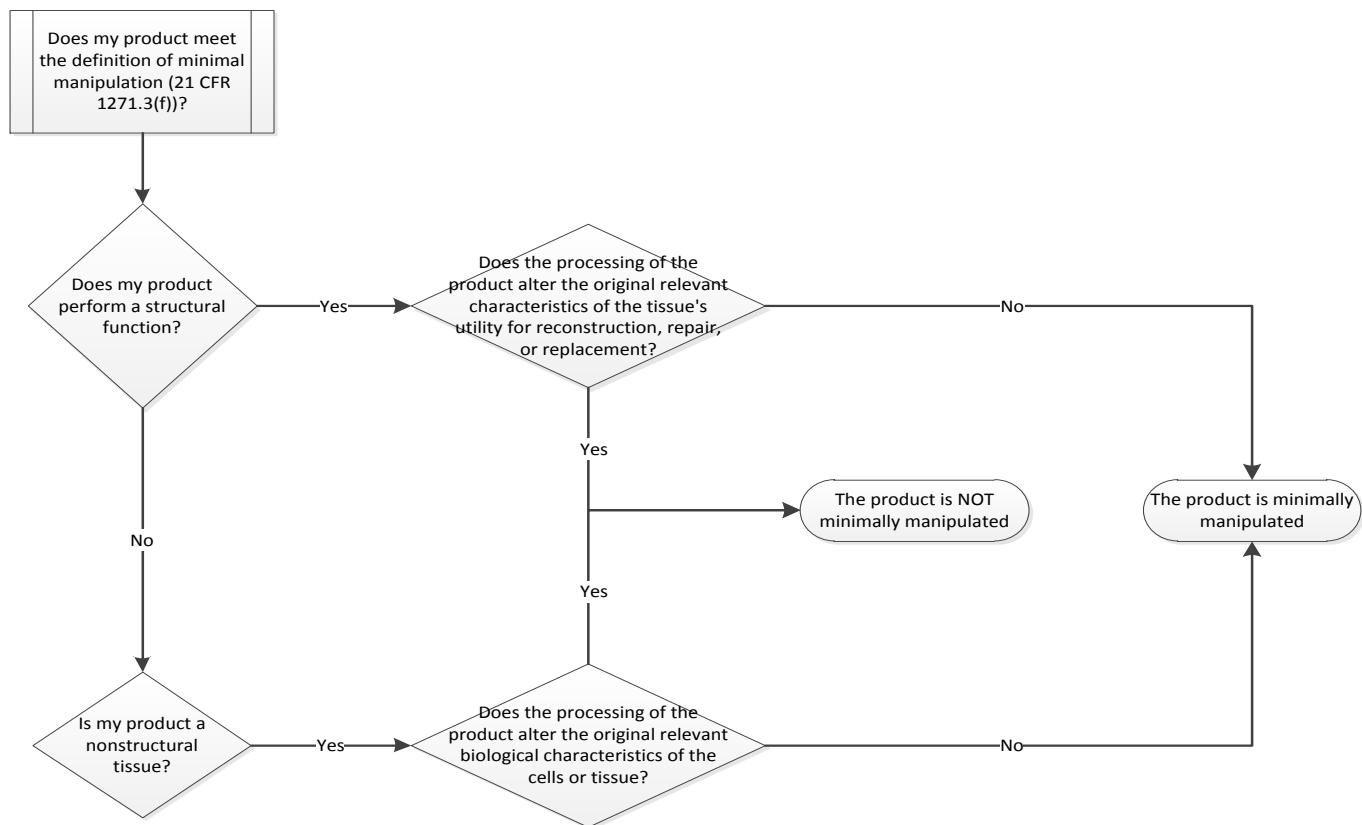
¹³⁰<http://www.ncbi.nlm.nih.gov/pubmed/9211605>

¹³¹<http://www.ncbi.nlm.nih.gov/pubmed/9211605>

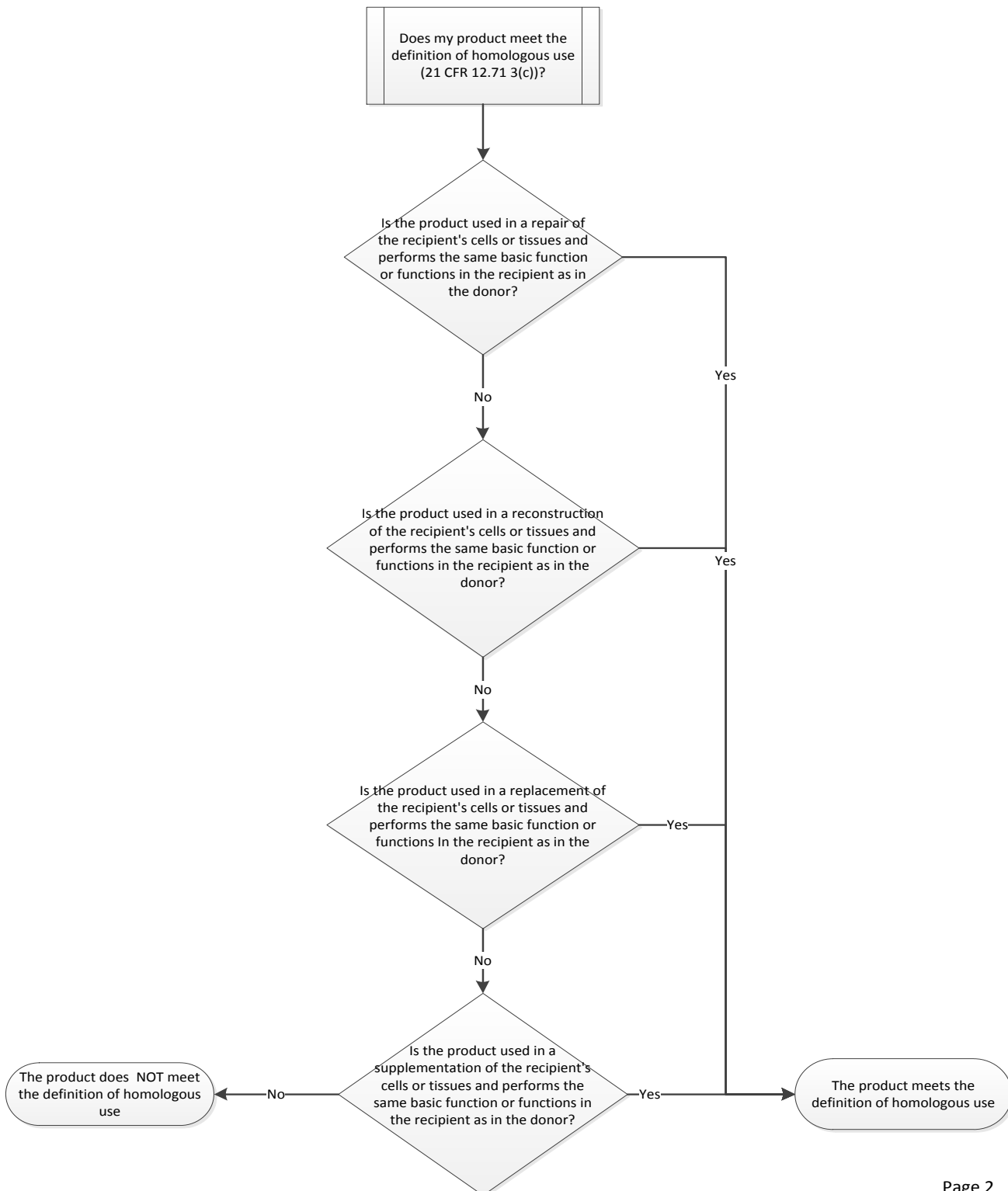
¹³²<http://www.umich.edu/~bme332/ch10ligten/bme332ligamenttendon.htm>

Appendix B: HCT/P Flow Diagrams

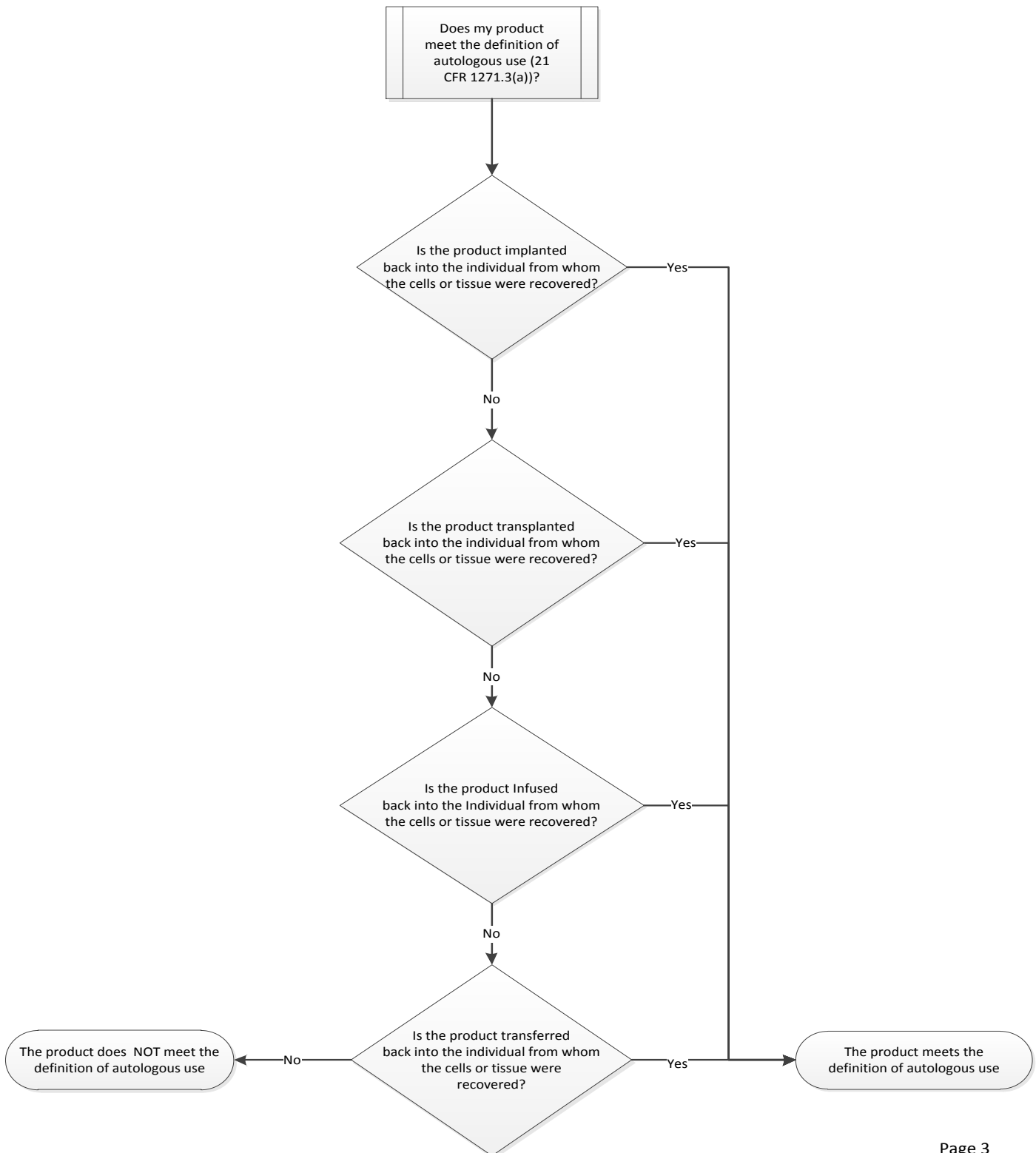
Does the product meet the Minimal Manipulation
criteria under 21 CFR 1271



Does the product meet the Homologous Use criteria under 21 CFR 1271



Does the product meet the Autologous Use criteria under 21 CFR 1271



Does the product meet the criteria to be regulated solely as a 361 HCT/P under 21 CFR 1271

